

Article

Joint Planning Method for Soft Open Points and Energy Storage in Hybrid Distribution Networks Based on Improved DC Power Flow

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Abstract

Intelligent soft open points (SOPs) and energy storage systems (ESSs) are effective ways to absorb distributed new energy in the spatial and temporal dimensions, and play an important role in improving the new-energy-carrying capacity of distribution networks. Existing planning models for SOPs and ESSs in distribution networks are often nonlinear and non-convex, and are usually transformed into a mixed-integer second-order cone optimization (MISOCP) model. However, this transformation often needs stringent relaxation conditions, and the solution speed and convergence performance of the model are poor. These disadvantages make traditional MISOCP models unsuitable for optimal planning for complex hybrid networks. To overcome these limitations, a joint planning method for AC/DC hybrid networks based on an improved DC power flow (IDCPF) algorithm is proposed in this paper. The proposed method transforms the original nonlinear model into an approximate linear model, improving the solution speed and accuracy of the model. The effectiveness of the proposed method is validated through case studies on an improved AC/DC 43-node network, which demonstrates the accuracy and numerical stability of the planning model.

Keywords: DC power flow; energy storage system; hybrid distribution network; joint planning; soft open point



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1. Introduction

As the penetration rate of new energy generation continues to increase, its intermittency and uncertainty pose significant challenges to the stable operation of power systems [1–3]. Therefore, it is imperative to effectively assess the new-energy-carrying capacity of existing power grids and utilize system flexibility resources to improve the proportion of new energy as well as the quality and reliability of the power supply.

Currently, to address the impacts brought about by distributed generation (DG) integration, numerous studies have focused on planning and regulation methods for high-penetration DG access. References [4–6] employ network reconfiguration strategies during distribution network planning to achieve DG configuration in high-penetration scenarios, while Ref. [7] develops a tripartite optimization model coordinating the interests of DG operators, distribution companies, and consumers through carbon/green certificate trading

mechanisms. The model incorporates demand response programs and energy storage flexibility to improve DG utilization while reducing carbon footprint in power generation and transmission. To address active distribution network planning, Ref. [8] establishes a multi-agent framework involving distributed generator owners, load aggregators, and distribution network operators (DNOs). This hierarchical structure features a bi-level optimization where DNOs optimize network expansion at the upper level while other stakeholders coordinate DG configuration at the lower level. Building on this, Ref. [9] proposes a multi-stage expansion planning framework with a two-layer optimization model. The upper layer determines optimal DG allocation schemes across planning phases, while the lower layer optimizes dispatch under various scenarios. From a power electronics perspective, Ref. [10] presents a smart inverter control strategy for PV-battery systems, implementing optimized Volt/Var regulation to enhance DG hosting capacity through reactive power compensation and optimal siting of distributed resources. Most of the aforementioned studies improve the new-energy-carrying capacity of the distribution network and enhance power quality by establishing multi-layer optimization models that include both DG planning and distribution network optimization stages. However, the proposed methods mainly focus on the temporal dimension for planning and configuring flexibility resources, without considering the influence of the distribution network's interconnection structure and spatial power flow distribution on DG integration capacity, which limit the effectiveness of improving the new-energy-carrying capacity.

The soft open point (SOP) is a power electronic device capable of replacing traditional interconnection switches, enabling flexible interconnection between different feeders in distribution networks and optimizing power flow distribution in the spatial dimension [11–13]. This has led to extensive research on planning and configuration methods for flexible interconnection distribution networks based on SOPs. Ref. [14] proposes a two-level coordinated planning model for DG and SOPs, leveraging the characteristics of unbalanced distribution networks. A multi-objective stochastic information gap decision model to minimize the comprehensive cost for distributed system operators (DSOs) in SOP planning is developed to address the cost uncertainties in [15]. Ref. [16] employs a two-stage robust optimization approach to evaluate photovoltaic (PV) carrying capacity in scenarios integrating SOPs and electric vehicles (EVs), enhancing solution efficiency through convex relaxation and column-and-constraint generation algorithms.

In joint planning, most studies involve DG and energy storage systems (ESSs); thus, the complementary spatial and temporal characteristics of ESSs and SOPs in power transmission have emerged as a research focus. Ref. [17] introduces a multi-port SOP and ESS collaborative planning method, optimizing distribution network flexibility based on the branch current distribution and network loss sensitivity indices. The effectiveness of this method was validated through comparisons using the IEEE 33-node test system. An integrated planning approach for ESOP devices (ESSs and SOPs) to achieve economic and flexible operation of distribution networks was proposed in [18]. Additionally, some studies have explored SOP planning and configuration from the perspective of fault recovery and resilience enhancement in distribution networks. Ref. [19] proposes a collaborative planning model for SOPs and remote-controlled switches (RCSs) to improve distribution network resilience. This model incorporates multiple fault recovery stages and optimized configurations to minimize equipment investment costs and load disconnection penalties, with its effectiveness verified through case studies.

However, most existing studies focus primarily on economic aspects, neglecting the impact of ESS and SOP joint planning on key indicators such as the distributed generation carrying capacity, power quality, supply capability, and reliability of distribution networks. Furthermore, multi-objective joint planning of ESSs and SOPs remains underexplored.

The aforementioned methods exhibit certain limitations. Second-order cone relaxation methods require carefully designed objective functions to drive the relaxation process toward optimal solutions. Therefore, these methods impose stringent requirements on relaxation conditions, limiting their general applicability and convergence performance. As the distribution network scale increases, the relaxation of numerous constraints significantly affects model solution speed, rendering these methods unsuitable for rapid assessment of renewable energy systems.

The DC power flow model simplifies power flow calculations by ignoring the relationship between reactive power and voltage, thereby reducing the computational burden of the model. Currently, extensive research is being conducted on the application scope of the DC power flow model and the factors affecting its solution accuracy. Ref. [20] investigates the impact of factors such as line series resistance, line impedance ratio, and voltage phase angle difference at both ends of the line on the accuracy of DC power flow. Ref. [21] proposes an extended DC power flow (EDCPF) model incorporating voltage magnitude, based on the quasi-linear relationship between voltage angle and voltage magnitude. The reliability and accuracy of the EDCPF model were validated through standard test systems. Ref. [22] develops a sensitivity analysis-based power flow (SAPF) method for multi-terminal high-voltage DC systems based on voltage source converters (VSCs). This method calculates state variables of extended AC grids through sensitivity analysis, significantly reducing computational burden and avoiding convergence issues in traditional sequential methods.

In summary, a clear research gap remains in the current literature. Although existing works have explored SOP and ESS integration, they predominantly focus on economic objectives and rely heavily on nonlinear or mixed-integer second-order cone optimization (MISOCP) models that suffer from stringent relaxation conditions and high computational burdens. There is a critical lack of a comprehensive, multi-objective joint planning framework that concurrently optimizes spatial-temporal flexibility, renewable hosting capacity, and power quality, while fundamentally overcoming the mathematical bottlenecks of traditional AC power flow in large-scale hybrid networks.

To achieve the joint planning of SOPs and ESSs in AC/DC hybrid networks, this paper proposes a bi-level optimization model for the flexible interconnection distribution network based on an improved DC power flow algorithm. The power flow constraints in this model are approximate linear equations, which avoids the cone relaxation of large-scale nonlinear constraints. This eliminates convergence issues while improving solution efficiency. Finally, the proposed model is validated using the improved AC/DC 43-node system.

The remainder of this paper is organized as follows: Section 2 details the proposed improved DC power flow (IDCPF) algorithm and formulates the multi-objective bi-level joint planning model for SOPs and ESSs. Section 3 presents the simulation results and comparative analysis. Finally, Section 4 concludes the paper.

2. Joint Planning Method for Hybrid Networks Based on Improved DC Power Flow Algorithm

2.1. Improved DC Power Flow Algorithm

In high-voltage power transmission networks, the DC power flow algorithm is commonly used to quickly calculate the power flow. However, in medium- and low-voltage distribution networks, the traditional DC power flow algorithm is no longer applicable. Therefore, an improved DC power flow (IDCPF) algorithm is introduced in this paper first. The improved algorithm is demonstrated by the following distribution network.

As it shows in Figure 1, in the analysis of the above radial distribution networks, the active and reactive power of each. This process is based on the power injection at each

node and the network's topology, while disregarding the active and reactive power losses in the lines.

$$\begin{cases} P_{ij} = P_{jj} + \sum_{k \in B_j} P_{jk} \\ Q_{ij} = Q_{jj} + \sum_{k \in B_j} Q_{jk} \end{cases} \quad (1)$$

where P_{ij} and Q_{ij} are the active and reactive power flows through branch ij , and P_{jj} and Q_{jj} are the respect power injections at node j . B_j denotes the child sets of node j . Thus, the active and reactive power of each branch can be recursively calculated using Equation (1). Based on the power flow distribution of the branches, the voltage phase angle at each node can be back-calculated using the following linear equation:

$$\begin{cases} \theta_j = \theta_i - (x_{ij}P_{ij} - r_{ij}Q_{ij}) \\ \theta_0 = 0 \end{cases} \quad (2)$$

where θ_i and θ_j represent the voltage phase angles at nodes i and j , respectively. Next, to calculate the voltage magnitude at each node, the complex power flow equation for an N -node distribution network is established as follows:

$$\bar{S} = \text{diag}[\bar{V}] \cdot \bar{Y}^* \cdot \bar{V}^* \quad (3)$$

where $\bar{S}$, $\bar{Y}$ and $\bar{V}$ represent the complex power injection vector, the complex admittance matrix, and the complex voltage vector of the distribution network nodes, respectively. $\text{Diag}[\bar{V}]$ denotes a diagonal matrix with elements from the complex vector $\bar{V}$ on its diagonal. Additionally, the complex voltage at node i can be expressed in the following form:

$$\bar{V}_i = (V_i^b + \Delta V_i) e^{j\theta_i} \quad (4)$$

where the real number V_i^b represents the reference value of the voltage magnitude, and the real number ΔV_i represents its correction term. Substituting Equation (4) into Equation (3) yields the following form of the power flow equation:

$$\bar{S} = \text{diag}[\mathbf{V}^b + \Delta \mathbf{V}] \cdot [\bar{Y}^* * \boldsymbol{\theta}] \cdot [\mathbf{V}^b + \Delta \mathbf{V}] \quad (5)$$

where $\boldsymbol{\theta}$ is an N -order complex matrix of phase angle differences, in which the elements $\theta_{ij} = e^{j(\theta_i - \theta_j)}$. The real vectors $\mathbf{V}^b$ and $\Delta \mathbf{V}$ represent the reference voltage vector and the correction vector, respectively. Let $\bar{Y}^+ = \bar{Y}^* * \boldsymbol{\theta}$; then, Equation (5) can be expanded as follows:

$$\begin{aligned} \bar{S} = & \text{diag}[\mathbf{V}^b] \cdot \bar{Y}^+ \cdot \Delta \mathbf{V} + \text{diag}[\Delta \mathbf{V}] \cdot \bar{Y}^+ \cdot \Delta \mathbf{V} \\ & + \text{diag}[\mathbf{V}^b] \cdot \bar{Y}^+ \cdot \mathbf{V}^b + \text{diag}[\Delta \mathbf{V}] \cdot \bar{Y}^+ \cdot \mathbf{V}^b \end{aligned} \quad (6)$$

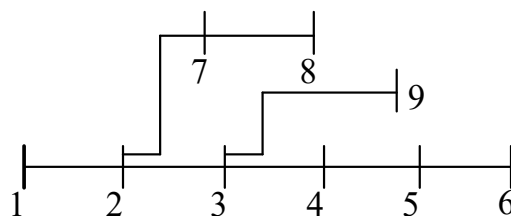


Figure 1. Topology of a 9-node radial distribution network.

Assuming ΔV is small and ignoring the higher-order terms, Equation (6) can be further simplified to:

$$\begin{cases} \Delta \bar{S} = \text{diag}[\Delta \mathbf{V}] \cdot \bar{\mathbf{Y}}^+ \cdot \mathbf{V}^b + \text{diag}[\mathbf{V}^b] \cdot \bar{\mathbf{Y}}^+ \cdot \Delta \mathbf{V} \\ \Delta \bar{S} = \bar{S} - \text{diag}[\mathbf{V}^b] \cdot \bar{\mathbf{Y}}^+ \cdot \mathbf{V}^b \end{cases} \quad (7)$$

$$\text{diag}[\bar{\mathbf{Y}}^+ \mathbf{V}^b] \cdot \Delta \mathbf{V} = \text{diag}[\Delta \mathbf{V}] \cdot \bar{\mathbf{Y}}^+ \cdot \mathbf{V}^b \quad (8)$$

Substituting Equation (8) into Equation (7) yields the following form of the power flow equation:

$$[\Delta \bar{S}]_{PQ} = \bar{A}_s \cdot [\Delta \mathbf{V}]_{PQ} \quad (9)$$

where $[\bullet]_{PQ}$ represents the matrix obtained after eliminating the data for PV nodes and slack nodes. The complex matrix $\bar{A}_s$ is given by $\text{diag}[\bar{\mathbf{Y}}^+ \mathbf{V}^b] + \text{diag}[\mathbf{V}^b] \cdot \bar{\mathbf{Y}}^+$, which is also obtained after eliminating the data for PV nodes and slack nodes. Separating the real and imaginary parts of the above complex matrix yields the following equations:

$$\begin{cases} [\Delta \bar{S}]_{PQ} = [\Delta \mathbf{P}]_{PQ} + j[\Delta \mathbf{Q}]_{PQ} \\ \bar{A}_s = \mathbf{A}_P + j\mathbf{A}_Q \end{cases} \quad (10)$$

Substituting Equation (10) into Equation (9) gives:

$$\begin{cases} [\Delta \bar{S}]_{PQ} = \bar{A}_s \cdot [\Delta \mathbf{V}]_{PQ} \\ [\Delta \mathbf{P}]_{PQ} = \mathbf{A}_P \cdot [\Delta \mathbf{V}]_{PQ} \\ [\Delta \mathbf{Q}]_{PQ} = \mathbf{A}_Q \cdot [\Delta \mathbf{V}]_{PQ} \end{cases} \quad (11)$$

Finally, the active and reactive power equations in Equation (11) can be combined into an overdetermined system of equations, and the least squares solution for this overdetermined system is as follows:

$$[\Delta \mathbf{V}]_{PQ} = \left([\mathbf{A}_P^T \ \mathbf{A}_Q^T] \cdot \begin{bmatrix} \mathbf{A}_P \\ \mathbf{A}_Q \end{bmatrix} \right)^{-1} \cdot \left([\mathbf{A}_P^T \ \mathbf{A}_Q^T] \cdot \begin{bmatrix} [\Delta \mathbf{P}]_{PQ} \\ [\Delta \mathbf{Q}]_{PQ} \end{bmatrix} \right) \quad (12)$$

This solution is the least squares format (LSQ) for solving the voltage magnitude. After calculating the voltage correction terms at the PQ nodes using the above formats, the voltage magnitude and phase angle information for all nodes in the entire distribution network can be obtained.

2.2. Joint Planning Model Based on the IDCPF Algorithm

The optimal allocation of DG, SOP, and ESS within distribution networks presents itself as a multi-objective optimization challenge that necessitates balancing the interests of multiple stakeholders. In order to holistically take into account elements like the economic benefit of the distribution network, renewable energy accommodation capacity, power supply proficiency, and power quality when arranging ESSs and SOPs, this study devises a multi-objective bi-level joint planning model for an SOP and ESS.

2.2.1. Upper-Level Planning Model

Objective Function 1: Optimal Economic Benefit.

The economic benefit of the distribution network is primarily composed of three main components: revenue from electricity sales, operational costs of the distribution network, and investment costs associated with the ESS and SOP. The optimization objective is to maximize the revenue of the distribution network. The operational costs of the distribution network are primarily attributed to three factors: the cost of electricity exchange between

the distribution network and the upstream grid, the operational costs of DG, and the network loss costs. These components are mathematically represented in Equation (13):

$$\min f_1 = c_S * S_{SOP} + c_E * S_{ESS} + C^G - C^S + C^{DG} + C^{LOSS} \quad (13)$$

where C^G represents the cost incurred from energy exchange between the distribution network and the upper-level grid. C^S is the revenue from electricity sales to users by the distribution network. C^{DG} denotes the cost of purchasing electricity from DG by the distribution network. C^{LOSS} refers to the cost of network losses within the distribution network. c_S is the annualized cost per unit capacity of SOP equipment over its entire lifecycle, and S_{SOP} is the installed capacity of the SOP within the distribution network. c_E represents the annualized cost per unit capacity of energy storage equipment over its entire lifecycle, and S_{ESS} is the total installed capacity of energy storage within the distribution network.

The annualized cost over the entire lifecycle for each piece of equipment can be calculated from its net present value (NPV) cost, expressed as follows:

$$\begin{cases} c_E = \zeta(r, l)[C_{E,init} + C_{E,rep} - C_{E,sal} + C_{E,om}] \\ c_S = \zeta(r, l)[C_{S,init} + C_{S,om}] \end{cases} \quad (14)$$

where $C_{E,init}$ and $C_{S,init}$ represent the initial net present value (NPV) cost of the unit capacity for energy storage and the SOP, respectively. $C_{E,om}$ and $C_{S,om}$ represent the annual operation and maintenance NPV cost of the unit capacity for energy storage and the SOP, respectively. $C_{E,rep}$ and $C_{E,sal}$ refer to the replacement cost and salvage income of the unit capacity for energy storage, respectively. $\zeta(r, l)$ is the capital recovery factor, where r is the actual discount rate and l is the equipment's service life. The formula for calculating the capital recovery factor is shown below:

$$\zeta(r, l) = \frac{T}{365} \frac{r(1+r)^l}{(1+r)^l - 1} \quad (15)$$

In this formula, T represents the number of daily sampling points within the optimization period. The cost incurred from energy exchange between the distribution network and the upper-level grid can be expressed by Equation (16):

$$C^G = \sum_{t=1}^T \lambda_t^G P_{ex,t} \Delta t \quad (16)$$

where λ_t^G is the price at which the distribution network purchases or sells electricity from or to the upper-level grid during time period t . $P_{ex,t}$ represents the exchange power between the distribution network and the upper-level grid during time period t . The cost C^{DG} of purchasing electricity from distributed generation by the distribution network can be expressed as follows:

$$C^{DG} = \sum_{i=1}^{N_{DG}} \sum_{t=1}^T \lambda_{i,t}^{DG} P_{i,t}^{DG} \Delta t \quad (17)$$

In this formula, N_{DG} is the number of grid-connected distributed generators in the system. $\lambda_{i,t}^{DG}$ represents the feed-in tariff of the i -th grid-connected DG during time period t . $P_{i,t}^{DG}$ denotes the output power of the i -th grid-connected DG during time period t .

Objective Function 2: Maximum Renewable Energy Hosting Capacity.

The maximum renewable energy hosting capacity is defined as the sum of the capacities of distributed energy sources connected to each node in the distribution network, with

the objective of maximizing the installed capacity of distributed energy sources. This value, denoted as S_{DG} , can be calculated using the following equation:

$$\min f_2 = -S_{DG} \quad (18)$$

Objective Function 3: Optimal Power Supply Capability.

The supply capability of the distribution network is a key indicator for evaluating the stable operation of the system. In this model, the system's supply capability is primarily assessed through two critical metrics: branch load margin and transformer load margin. To derive the optimal objective function for the system's supply capability, a normalization method is employed. The optimization goal is to minimize the weighted sum of the branch load margin and transformer load margin, as formulated in Equation (19).

$$\min f_3 = -(w_1 I^M + w_2 T^M) \quad (19)$$

In this formula, w_1 is the weighting coefficient for the branch load margin, I^M represents the branch load margin of the system, w_2 is the weighting coefficient for the transformer load margin, and T^M denotes the transformer load margin of the system, with $w_1 + w_2 = 1$.

The branch load margin, as defined in the equation above, represents the ratio of the available capacity of a branch in the distribution network to its maximum line capacity. A higher branch load margin indicates that the available capacity of the branch is sufficient to meet the actual load demand. Conversely, a lower margin suggests that the branch may be at risk of overloading. The specific calculation method is as follows:

$$I^M = \frac{1}{TN_L} \sum_{t=1}^T \sum_{ij \in A_L \cup D_L} (S_{lmax} - S_{ij,t}) / S_{lmax} \quad (20)$$

where A_L and D_L represent the sets of AC lines and DC lines, respectively. N_L is the total number of branches in the system. S_{lmax} is the maximum capacity limit of branch ij , and $S_{ij,t}$ is the apparent power of branch ij at time t . The concept of transformer load margin in the distribution network is similar to that of the branch load margin. It is the ratio between the transformer's rated capacity and the actual load. The calculation method is shown in Equation (21).

$$T^M = \frac{1}{T} \sum_{t=1}^T \frac{S_{max}^T - S_t^T}{S_{max}^T} \quad (21)$$

where S_{max}^T represents the rated capacity of the transformer at the substation node in the distribution system, and S_t^T denotes the apparent power of the transformer at time t .

Objective Function 4: Optimal Power Quality of the System.

Voltage deviation rate is a key indicator for assessing the power quality in a distribution network. The comprehensive voltage deviation rate of the distribution network can represent this quality. The optimization goal is to minimize the voltage deviation rate, as shown in Equation (22).

$$\min f_4 = \frac{1}{TN_D} \sum_{t=1}^T \sum_{i \in A_N \cup D_N} (U_{i,t} - U_{i,ref})^2 \quad (22)$$

where A_N and D_N represent the sets of AC nodes and DC nodes, respectively. N_D is the total number of nodes in the system, and $U_{i,t}$ denotes the voltage value at node i during time period t . $U_{i,ref}$ represents the rated voltage at node i .

Constraints:

$$0 \leq S_{DG} \leq \sum_{i \in A_N \cup D_N} \alpha_i S_{DG\max} \quad (23)$$

$$0 \leq S_{ESS} \leq \sum_{i \in A_N \cup D_N} \beta_i ESS_{\max} \quad (24)$$

$$\sum_{i \in A_N \cup D_N} \alpha_i = N_{DG} \quad (25)$$

$$\sum_{i \in A_N \cup D_N} \beta_i = N_{ESS} \quad (26)$$

In these formulas, $S_{DG\max}$ represents the upper limit of the capacity of DGs that can be connected to each node, and $ESS_{\max}$ represents the upper limit of the capacity of ESSs configured at each node, both of which are predetermined values. α_i and β_i represent the connection status, which are 0–1 variables: a value of 0 indicates that the node does not connect to a distributed generator or energy storage, while a value of 1 indicates a connection. N_{DG} denotes the allowed number of DG connections in the system, and N_{ESS} denotes the allowed number of ESS connections in the system, both of which are predetermined values.

2.2.2. Lower-Level Operation Model

Objective Function: Sum of network losses and SOP losses.

The lower-level operational model focuses on identifying the ideal operational status of the distribution network. This is achieved by reducing the combined total of network losses and SOP losses, following the determination of the total installed capacity and locations of photovoltaic and energy storage devices. The objective function is articulated as follows:

$$\min f = \sum_{t=1}^T \sum_{ij \in A_L \cup D_L} R_{ij} I_{ij,t}^2 + \sum_{t=1}^T \sum_{i=1}^{N_{SOP}} P_{i,t}^{SOP,loss} \quad (27)$$

where R_{ij} represents the resistance of the corresponding line, N_{SOP} is the number of SOPs connected to the distribution network, and $P_{i,t}^{SOP,loss}$ denotes the active power loss of the i -th SOP during time period t .

Constraints:

(1) SOP Operating Constraints

The soft open point aims to replace traditional feeder tie switches with controllable power electronic converters, thereby achieving flexible and soft connections between feeders under normal conditions. This allows for flexible, fast, and precise power exchange control and power flow optimization [23–25]. In this paper, a back-to-back voltage source-type SOP is used, and the following relevant constraints are considered.

(a) SOP Active Power Balance Constraint

The active power output from the SOP should be equal to the active power input to the SOP plus the total power losses generated by the converters at both ports. Therefore, the SOP active power balance constraint can be described as follows:

$$P_{i,t}^{SOP} + P_{j,t}^{SOP} + P_{i,t}^{SOP,loss} + P_{j,t}^{SOP,loss} = 0 \quad (28)$$

where $P_{i,t}^{SOP}$ represents the active power transmitted by the SOP at the port connected to node i during time period t , and $P_{j,t}^{SOP}$ represents the active power transmitted by the SOP at the port connected to node j during time period t . $P_{i,t}^{SOP,loss}$ represents the converter power loss at the SOP port connected to node i during time period t , and $P_{j,t}^{SOP,loss}$

represents the converter power loss at the SOP port connected to node j during time period t .

(b) SOP Active Power Loss Constraint

The loss of each port converter of the SOP is proportional to the sum of the squares of the active and reactive power it transmits. The SOP active power loss constraint is described as follows:

$$P_{i,t}^{\text{SOP,loss}} = A_i^{\text{SOP}} \sqrt{(P_{i,t}^{\text{SOP}})^2 + (Q_{i,t}^{\text{SOP}})^2} \quad (29)$$

$$P_{j,t}^{\text{SOP,loss}} = A_j^{\text{SOP}} \sqrt{(P_{j,t}^{\text{SOP}})^2 + (Q_{j,t}^{\text{SOP}})^2} \quad (30)$$

where A_i^{SOP} is the loss coefficient of the SOP converter connected at node i , and A_j^{SOP} is the loss coefficient of the SOP converter connected at node j . $Q_{i,t}^{\text{SOP}}$ represents the reactive power transmitted by the SOP at the port connected to node i during time period t , and $Q_{j,t}^{\text{SOP}}$ represents the reactive power transmitted by the SOP at the port connected to node j during time period t .

(c) SOP Capacity Limit Constraint

At any given time, the active and reactive power transmitted by the converters at each SOP port should be less than their rated capacity. The SOP capacity limit constraint is described as follows:

$$\sqrt{(P_{i,t}^{\text{SOP}})^2 + (Q_{i,t}^{\text{SOP}})^2} \leq S_i^{\text{SOP}} \quad (31)$$

$$\sqrt{(P_{j,t}^{\text{SOP}})^2 + (Q_{j,t}^{\text{SOP}})^2} \leq S_j^{\text{SOP}} \quad (32)$$

where S_i^{SOP} represents the capacity of the SOP converter connected at node i , and S_j^{SOP} represents the capacity of the SOP converter connected at node j .

(2) AC Line Power Flow and Voltage Constraints

(a) Power Flow Constrains

To avoid second-order cone relaxation, this paper does not employ the conventional branch power flow equations, but instead uses the improved DC power flow algorithm to describe the line power flow constraints.

$$\begin{cases} P_{ji,t} = \sum_{k \in B_i} P_{ik,t} + P_{i,t}^{\text{inj}} \\ Q_{ji,t} = \sum_{k \in B_i} Q_{ik,t} + Q_{i,t}^{\text{inj}} \end{cases} \quad \forall i \in B_j \cap A_N \quad (33)$$

$$\begin{cases} P_{i,t}^{\text{DG}} + P_{i,t}^{\text{SOP}} - P_{i,t}^{\text{load}} - P_{i,t}^{\text{bess}} - P_{i,t}^{\text{VSC,AC}} = P_{i,t}^{\text{inj}} \\ Q_{i,t}^{\text{SOP}} - Q_{i,t}^{\text{load}} - Q_{i,t}^{\text{VSC,AC}} = Q_{i,t}^{\text{inj}} \end{cases} \quad (34)$$

where $P_{ij,t}$ and $Q_{ij,t}$ represent the active and reactive power flowing through branch ij at time t , respectively. $P_{i,t}^{\text{inj}}$ and $Q_{i,t}^{\text{inj}}$ are the active and reactive power injected into node i , respectively. $P_{i,t}^{\text{DG}}$ and $P_{i,t}^{\text{load}}$ are the active power of distributed renewable energy and the load at node i at time t , respectively. $Q_{i,t}^{\text{load}}$ is the reactive power of the load at node i at time t . $P_{i,t}^{\text{bess}}$ is the output power of the ESS at node i at time t . $P_{i,t}^{\text{VSC,AC}}$ and $Q_{i,t}^{\text{VSC,AC}}$ are the active and reactive power injected into the AC side of the voltage source converter (VSC) at time t , respectively.

(b) Node Voltage Constrains

$$\begin{cases} \Delta \mathbf{U} = \mathbf{A}_s^{-1} \cdot \Delta \mathbf{S} \\ \Delta S_{i,t} = P_{i,t}^{\text{inj}} + jQ_{i,t}^{\text{inj}} - \sum_{j \in \mathbf{B}_i} y_{ij}^+ U_i^b U_j^b \quad \forall i \in \mathbf{A}_N \\ V_{l,A} \leq \Delta U_{i,t} + U_i^b \leq V_{u,A} \end{cases} \quad (35)$$

where $V_{l,A}$ and $V_{u,A}$ are the lower and upper limits of AC line voltage. U_i^b represents the reference value of the voltage magnitude of node i , and $\Delta U_{i,t}$ represents its correction term at time t . The remaining parameters have the same meaning as above.

(3) DC Line Power Flow and Voltage Constraints

(a) Power Flow Constrains

Similar to the AC lines, the power flow constraints for the DC lines can also be obtained.

$$\begin{cases} P_{ji,t} = \sum_{k \in \mathbf{B}_i} P_{ik,t} + P_{i,t}^{\text{inj}} \\ P_{i,t}^{\text{DG}} - P_{i,t}^{\text{load}} - P_{i,t}^{\text{bess}} + P_{i,t}^{\text{VSC,DC}} = P_{i,t}^{\text{inj}} \end{cases} \quad \forall i \in \mathbf{B}_i \cap \mathbf{D}_N \quad (36)$$

where $P_{i,t}^{\text{VSC,DC}}$ is the active power injected into the DC side of the VSC at time t . The remaining parameters have the same meaning as above.

(b) Node Voltage Constrains

$$\begin{cases} \mathbf{U} = \mathbf{B}_s^{-1} \cdot \mathbf{P}_L \\ V_{l,D}^2 \leq U_{i,t} \leq V_{u,D}^2 \end{cases} \quad \forall i \in \mathbf{D}_N \quad (37)$$

where $\mathbf{U}$ represents the vector of the square of the amplitude of the DC node voltage, and $\mathbf{P}_L$ represents the DC line power vector. The matrix $\mathbf{B}_s$ is determined by the system admittance matrix. $V_{l,D}$ and $V_{u,D}$ are the lower and upper limits of DC line voltage.

(4) Energy Storage System Operating Constraints

(a) Energy Storage Capacity Constraint

Considering the impact of the State of Charge (SOC) on the battery's lifespan, it is necessary to set upper and lower limits on the real-time capacity of the energy storage system, which can be expressed as:

$$\begin{cases} ESS_{i,t+\Delta t} = ESS_{i,t} + k_{bi} P_{i,t}^{\text{bess-in}} \Delta t + P_{i,t}^{\text{bess-out}} \Delta t / k_{bo} \\ \eta_{\min} ESS_i \leq ESS_{i,t} \leq \eta_{\max} ESS_i \\ P_{i,t}^{\text{bess}} = P_{i,t}^{\text{bess-in}} + P_{i,t}^{\text{bess-out}} \end{cases} \quad (38)$$

In this formula, $ESS_{i,t}$ and $ESS_{i,t+\Delta t}$ represent the remaining energy storage capacity at node i at times t and $t + \Delta t$, respectively. $\eta_{\min}$ and $\eta_{\max}$ are the minimum and maximum values of the State of Charge (SOC), respectively. ESS_i represents the energy storage capacity configured at node i . $P_{i,t}^{\text{bess-in}}$ and $P_{i,t}^{\text{bess-out}}$ are the charging and discharging power of the energy storage at node i at time t , respectively. k_{bi} and k_{bo} represent the charging and discharging efficiency of the energy storage, respectively.

(b) Energy Storage Charging and Discharging Power Constraint

Since the energy storage batteries are connected to the grid through converters, the charging and discharging power of the energy storage system is influenced not only by the

charging and discharging rate of each battery but also by the capacity of the grid-connected converter. The constraint is as follows:

$$\begin{cases} 0 \leq p_{i,t}^{\text{bess-in}} \leq U_{i,ch,t} p_{i,ch}^{\text{max}} \\ U_{i,dis,t} p_{i,dis}^{\text{max}} \leq p_{i,t}^{\text{bess-out}} \leq 0 \\ U_{i,ch,t} + U_{i,dis,t} \leq 1 \end{cases} \quad (39)$$

where $U_{i,ch,t}$ and $U_{i,dis,t}$ represent the charging and discharging states of the energy storage at node i at time t , respectively. $U_{i,ch,t} = 1$ indicates that the energy storage is charging at that time, while $U_{i,dis,t} = 1$ indicates that it is discharging. $p_{i,ch}^{\text{max}}$ and $p_{i,dis}^{\text{max}}$ are the maximum charging and discharging power limits of the energy storage at node i , respectively.

(5) VSC Operating Constraints

The VSC serves as the connection point for the AC-DC lines, and its constraints include capacity constraints and line power constraints.

$$\begin{cases} (p_{i,t}^{\text{VSC,AC}})^2 + (q_{i,t}^{\text{VSC,AC}})^2 \leq S_{i,\text{VSC}}^2 \\ p_{i,t}^{\text{VSC,AC}} + p_{i,t}^{\text{VSC,DC}} = 0 \end{cases} \quad (40)$$

where $S_{i,\text{VSC}}$ represents the rated apparent power of the VSC at node i . The remaining parameters have the same meaning as above.

2.2.3. Solution for the Bi-Level Optimization Model

To streamline the solution process, an enhanced Multi-objective Particle Swarm Optimization (MOPSO) algorithm is used in the model. MOPSO is utilized to ascertain the upper-level DG and energy storage siting and sizing strategies, whereas the lower-level model is solved by the business solver Gurobi. The flowchart is depicted in Figure 2.

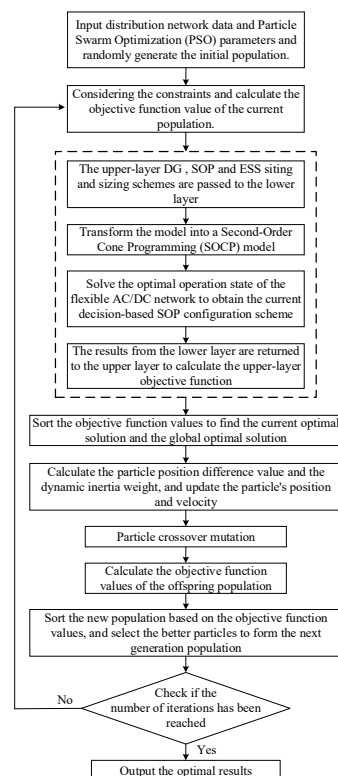


Figure 2. Hybrid algorithm flowchart.

3. Simulation Analysis and Algorithm Validation

3.1. Case Overview and Parameter Settings

This paper modifies the IEEE 33-node system into a 43-node AC/DC flexible interconnected distribution network, and verifies and analyzes the double-layer planning model and method proposed in the above text. This AC/DC distribution network has two DC lines, as shown in Figure 3, and the load conditions of each node on the DC lines are shown in Table 1. The rated voltage of the AC part is 12.66 kV, the rated voltage of the DC part is ± 10 kV, the capacity of the VSC is 500 kVA, the minimum installation capacity of the SOP is 100 kVA, the maximum installation capacity is 500 kVA, and the power loss coefficient of the converter at each end is 0.02. The capacity of the distribution network transformer is 10 MVA, and the maximum allowable reverse power does not exceed 80% of the transformer capacity. The remaining parameters are the same as the IEEE 33-node standard case.

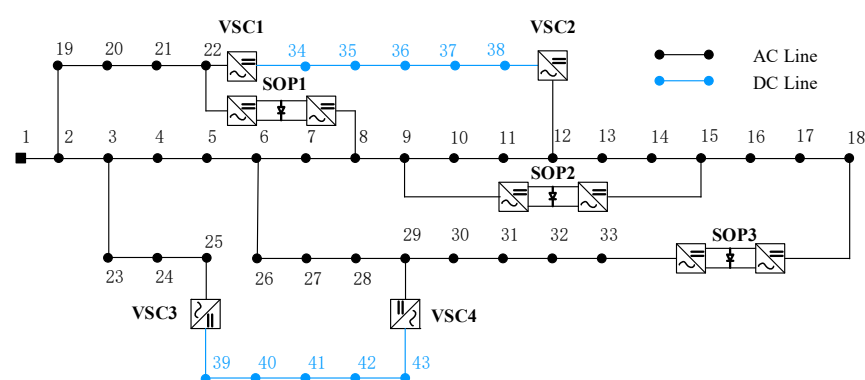


Figure 3. The topology of the improved 43-node AC/DC hybrid distribution network.

Table 1. Load on DC line.

Node	Load/kW	Node	Load/kW
34	40	39	40
35	120	40	140
36	80	41	60
37	50	42	70
38	150	43	100

Distributed power generation, exemplified by photovoltaic systems, involves 24 discrete sampling points within a single day, derived from a comprehensive dataset encompassing 8760 annual operational hours. The K-means algorithm is employed to perform clustering operations, extracting representative daily source-load unit capacity curves, as illustrated in Figure 4. Within the system, the node voltage is meticulously maintained within a per-unit range of 0.95 to 1.05, ensuring stable operational conditions. Additionally, the SOC for energy storage components is confined between 0.2 and 1.0 to optimize performance and longevity. Parameters pertinent to SOP and energy storage devices are comprehensively detailed in Table 2. The parameters for the Particle Swarm Optimization (PSO) algorithm are set as follows: the population size is 100, the number of iterations is 200, the mutation rate is 0.05, the crossover rate is 0.1, the minimum inertia coefficient $\omega_{\min}$ is 0.4, $\omega_{\max}$ is 0.9, and the dynamic learning factors are $c_{1k} = 2.5 + (0.5-2.5)k/\text{gen}$ and $c_{2k} = 0.5 + (2.5-0.5)k/\text{gen}$, where k is the current number of iterations and gen is the total number of iterations. The testing environment is the 12th Gen Intel(R) Core(TM) i7-12700H CPU with a main frequency of 2.70 GHz and 16 GB of memory, the development environment is Win 11 64-bit, and the simulation software is Matlab R2020b.

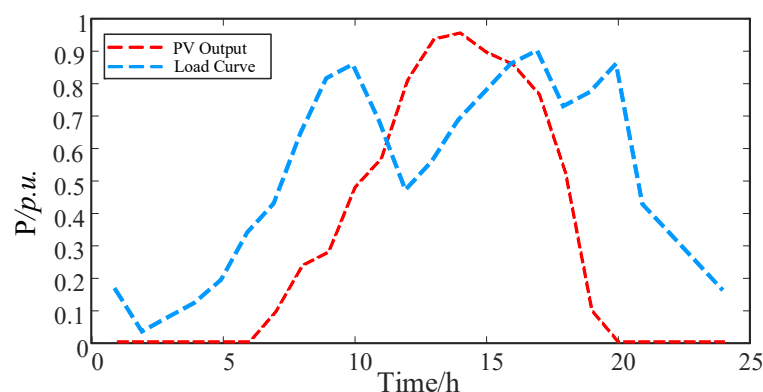


Figure 4. Typical-day DG output and load curve.

Table 2. SOP and storage device parameters.

Parameter	Value
Discount Rate	0.08
Economic Service Life/Years	20
SOP Unit Capacity Investment Cost/(Yuan/kVA)	1000
SOP Loss Coefficient	0.02
SOP Operation and Maintenance Coefficient	0.01
Energy Storage Capacity Cost/(Yuan/kWh)	1000
Energy Storage Power Cost/(Yuan/kW)	1500
Energy Storage Operation and Maintenance Cost/(Yuan/kWh)	0.0195

3.2. Planning Results and Analysis

Using the parameters and solution methods mentioned above, the multi-objective joint planning model for intelligent soft open points and energy storage in AC/DC flexible interconnected distribution networks is solved. Since the upper-level model is a multi-objective optimization problem, the solution is a set of Pareto optimal solutions, as depicted in Figure 5. To select the optimal solution from these Pareto optimal solutions, this study employs the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) method. This method compares each alternative solution with the positive and negative ideal solutions, and ultimately determines the optimal solution. The weight values of the indicators used in the TOPSIS method are provided in Table 3. Finally, the optimal planning schemes for SOP, ESS, and DG are presented in Table 4.

Table 3. Index weight value.

Category	f1	f2	f3	f4
Indicator Weights	0.316	0.344	0.24	0.1

Table 4. Configuration scheme of distribution network.

Category	Access Node	Access Capacity	Units
DG	2, 7, 21, 37, 39, 43	6, 1.33, 1.63, 0.79, 1.07, 0.77	MW
ESS	7, 31, 38	0.18, 1.0, 0.92	MWh
SOP	8–22, 9–15, 18–33	0.5, 0.25, 0.3	MVA

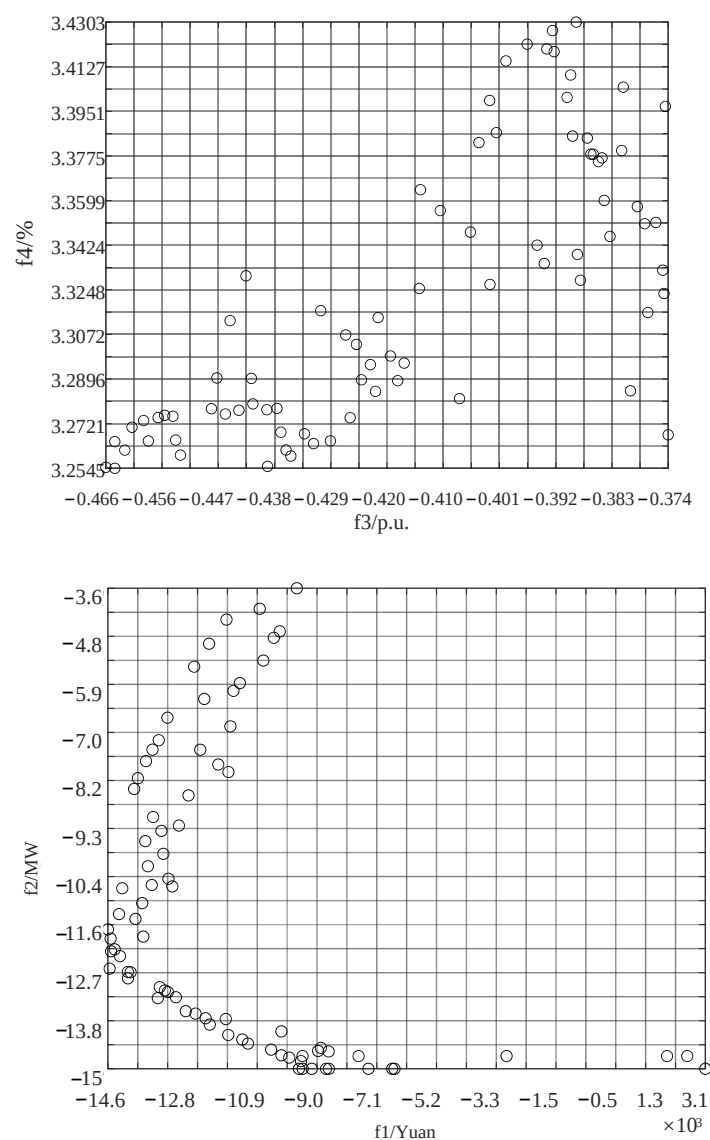


Figure 5. Pareto frontier.

To verify the effectiveness of the proposed multi-objective optimization method, this paper sets up the following three scenarios to compare the DG penetration rate and distribution network performance indicators under different scenarios.

Scenario I: Only the SOP is connected.

Scenario II: On the basis of SOP connection, a 2.1 MWh energy storage device is connected at a single point (Node 31).

Scenario III: The proposed joint plan.

The supremacy of the proposed method is concluded by the numerical values, which are shown in Table 5.

Table 5. Index comparison between different scenarios.

	f1/Yuan	f2/MW	f3/p.u.	f4/%
Scenario I	-11.1×10^3	8.9	-0.32	3.5
Scenario II	-12.8×10^3	10.4	-0.41	3.33
Scenario III	-13.5×10^3	11.59	-0.42	3.3

From Table 5, it can be seen that due to the additional energy storage equipment configured in Scenarios II and III, the access capacity of DG has significantly increased, resulting in a substantial reduction in the purchase cost of the distribution network. Although the one-time cost of purchasing energy storage is relatively high, the cost apportionments of the ESS still lead to an increase in the overall revenue. Additionally, with the increase in DG, although the random fluctuation increases, the SOP and ESS effectively improve the power quality of the distribution network. Further, the method proposed in this paper enables the SOP and ESS to optimize the line flow in both the spatial and temporal dimensions; thus, the capability to supply power will be better.

Figure 6 shows the active power transmission of SOP (9–15) and SOP (8–22) within a typical day, in which P denotes the active power measured in kilowatts (kW), and t represents time, measured in hours (h). The three curves in the figure represent different operation conditions under the aforementioned three scenarios in the flexible interconnected AC/DC distribution network.

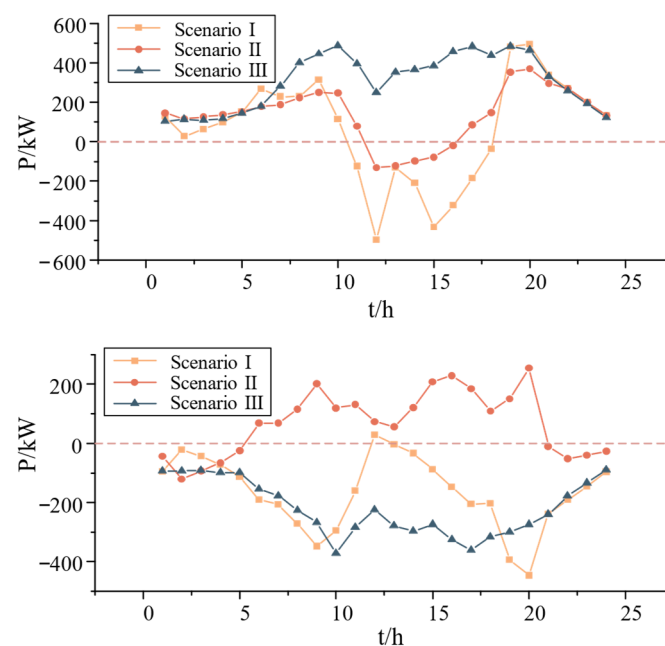


Figure 6. Active power transmission of SOPs under different scenarios.

In Scenario I, the absence of energy storage devices leads to a significant imbalance between generation and load across feeders. This results in marked fluctuations in the active power transmitted by the SOP, frequent power flow direction changes, and an intensified power flow imbalance, all of which negatively impact the stability of the distribution network. In Scenario II, despite the connection of energy storage devices, the reduction in power transmission fluctuation is limited. The energy storage devices are only connected at a single point and do not consider the complementary interaction between energy storage and the SOP, leading to a decrease in total transmitted power and an underutilization of the SOP's power transfer capability. Scenario III, however, takes into account the flexibility of dispersed energy storage connections and fully exploits the complementary effect between the SOP and the energy storage devices. This results in more stable power transmission and a higher total power output.

Figure 7 illustrates the magnitude of the charging and discharging power of the energy storage in the AC/DC flexible distribution network during a typical day in Scenario III.

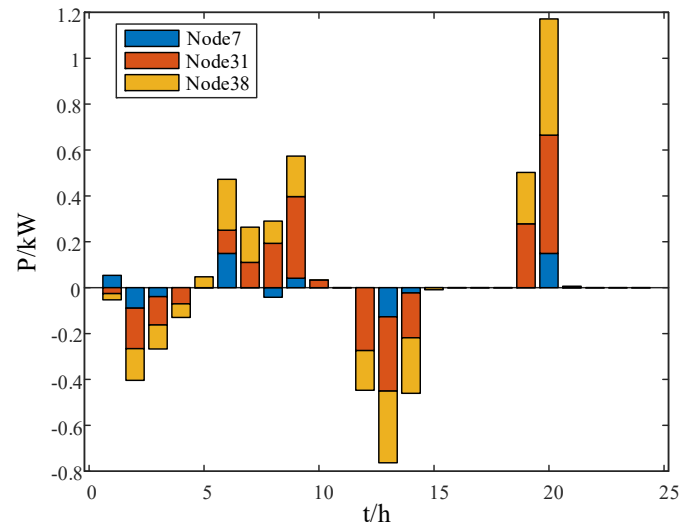


Figure 7. Energy storage charging and discharging power.

From Figure 7, it can be observed that the energy storage devices at each node mainly charge during the night-time load valley period (0–5 h) and the midday high photovoltaic generation period (11–14 h). On the one hand, charging during the load valley period allows for peak-shaving and valley-filling arbitrage. Additionally, the charging power during the midday is higher and highly coincides with the period of peak photovoltaic output, indicating that the energy storage batteries absorb the excess photovoltaic generation in the system during the middle of the day, achieving power transfer in the temporal dimension, smoothing the midday photovoltaic peak, and enhancing the distribution network's capacity to accommodate photovoltaic generation. At other times during the day, the energy storage focuses on discharging during the morning and evening load peak periods, smoothing the load peaks. At the same time, in conjunction with the spatial power transfer function of the SOP, it makes the power flow between feeders more balanced, improving the system's power supply capability and reducing network losses.

Figures 8–10 offer a comparison of voltage deviation across the entire day, voltage variations at the node with significant voltage fluctuations (Node 18), and line loading conditions during the midday peak photovoltaic generation period in the AC/DC flexible distribution network across the three discussed scenarios.

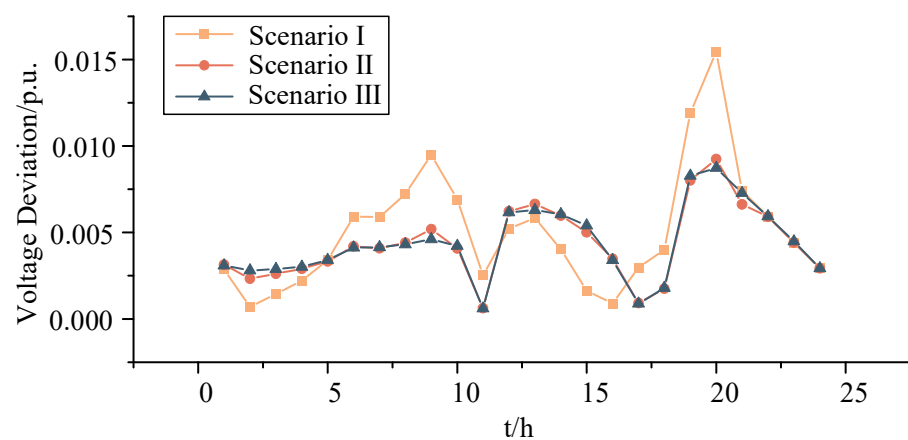


Figure 8. Voltage deviation under different scenarios.

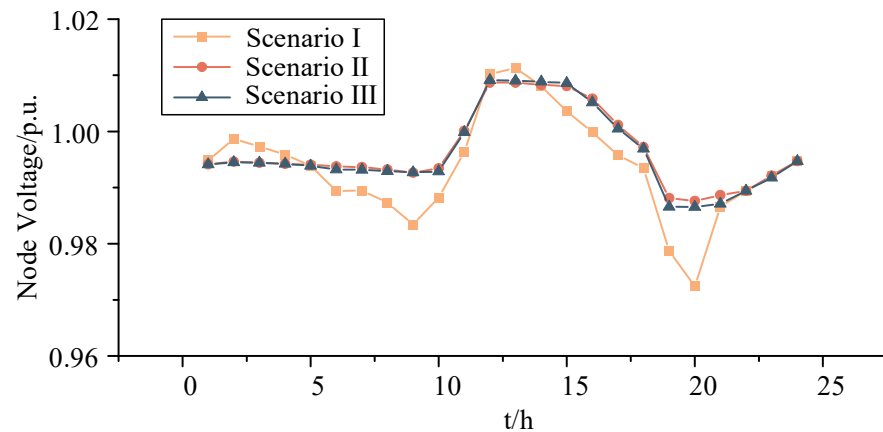


Figure 9. Voltage values under different scenarios.

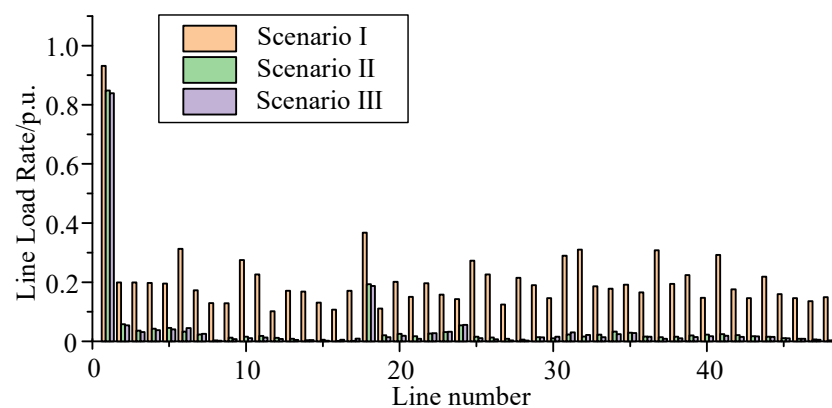


Figure 10. Line load rate comparison under different scenarios.

The results in Figures 8–10 demonstrate that the flexible AC/DC distribution network configuration, derived from the multi-objective planning method presented in this paper, exhibits several key advantages. The configuration shows enhanced complementarity among devices, ensuring more effective interaction between components. It also provides a planning scheme that is better suited to the existing distribution network structure, making it more practical for implementation. Furthermore, the configuration exhibits improved performance, leading to reduced system voltage deviation, mitigated voltage fluctuations, and optimized branch load distribution. These improvements collectively contribute to a more flexible and reliable system, better equipped to handle varying operational conditions and enhance the overall power supply quality.

Finally, in order to verify the solution performance of the IDCPF algorithm, this paper takes the solution time of the traditional SOCP method as the benchmark, and uses these two algorithms to solve the lower-level operation model within the three scenarios. The results are shown in Table 6.

Table 6. Solution time comparison between the two methods.

	SOCP/s	IDCPF/s
Scenario I	15.3	8.9
Scenario II	18.1	13.4
Scenario III	28.7	22.5

From Table 6, it can be seen that the traditional SOCP method takes more time to solve the model compared to the proposed IDCPF method. Specifically, in Scenario I,

due to a large number of quadratic constraints brought about by the second-order cone relaxation of the power flow equation, the solution time of SOCP is slower. Similarly, in Scenario II, only the decision variables for energy storage charging and discharging have been added; thus, the solution time of IDCPF is also faster. In Scenario III, in addition to the aforementioned variables, decision variables (integer variables) for the storage capacity at different locations are also introduced. In this scenario, although the IDCPF algorithm approximately linearizes the power flow equations, it still takes a considerable amount of time to handle the integer variables. Thus, the advantages of the proposed algorithm are not sufficiently prominent. It is worth noting that the solution time above is only for lower-level operation model; the locations of ESSs (0–1 variables) are given by the upper-level model. However, if the location is the decision variable, the solution time of Scenario III is 442.3 s. Thus, the 0–1 variables consume a lot of time during the solution process, which is the reason why the joint planning model in this paper is decomposed into a two-level model for solution.

4. Conclusions

This paper addresses the optimization configuration problem of various devices such as DG, energy storage, and SOP devices in flexible AC/DC distribution networks. A multi-objective particle swarm algorithm based on an improved DC power flow algorithm is used to solve the joint planning model. From the case study of the improved 43-node flexible AC/DC distribution network, it can be concluded that the multi-objective joint planning model not only considers the enhancement of the distribution network's new-energy-carrying capacity but also takes into account the economic operation of the distribution network and the improvement in the overall power quality of the system. Compared with the pre-planning scenario, the economic benefit, DG capacity, and power supply capability increased by 21.6%, 30.2%, and 31.25%, respectively. Additionally, the hybrid algorithm based on IDCPF can be applied to large-scale mixed-integer nonlinear programming problems. Combined with the optimal solution evaluation strategy, it can obtain the optimal configuration scheme for AC/DC flexible distribution networks. However, this research also faced certain challenges, primarily in balancing conflicting multi-objective dimensions (e.g., economic cost vs. power quality) and ensuring the numerical convergence of the IDCPF algorithm when scaling to complex network topologies.

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